23-May-24

To

M/s. PA KNIT COMPOSITE LIMITED,
MYMENSINGH,
DHAKA,
BANGLADESH.

Our Ref: Techno Commercial Proposal for ZLD based Effluent Treatment Plant.

[Proposal No: 11.May.2024/WTT-0685/BAN/RP-ETP/7.2M/R0.1/TCP].

Dear Sir,

WTT International Private Limited (WTT) is pleased to provide M/s. PA KNIT COMPOSITE LIMITED with the offer for supply of engineering & materials for ZLD based Effluent Treatment Plant of capacity 7200 M^3/Day. In developing this offer WTT worked with you in an effort to understand your project and business needs. The attached proposal outlines the solutions we feel will best meet these objectives.

We greatly appreciate your consideration of WTT for this project. Our measure of success is how well we deliver solutions that help our client to meet their critical business objectives. We hope to have the opportunity to demonstrate this with your good selves.

Yours Sincerely,

D. Venkatesh
Managing Director

Proposal for:

1) ZLD based Effluent Treatment Plant of capacity 7200 M^3/Day.

Proposal No: 11.May.2024/WTT-0685/BAN/RP-ETP/7.2M/R0.1/TCP

S.NO	DESCRIPTION	DATE	PREPARED	APPROVED
1.	Techno Commercial Proposal – "ZLD based Effluent Treatment Plant".	23-May-24	SK - GET AR - AGM	DV - MD

CONFIDENTIALITY

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WE FEED DELIGHTED TO PRESENT OUR PATENTED SYSTEMS

Award For Excellence in Water Management

CII (Confederation of Indian Industry) *awarded*
WTT International *with the prestigious*

"Excellence in Water Management Award"

recognizing the advanced water treatment technologies
innovated by ***WTT International***
As a part of Industrial Water / Waste Management competition 2023

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1. Technical And Engineering Details

1.1. Basis of Design

- ✓ *Rotary Brush Screener system proposed to remove coarse particles from the Effluent water*
- ✓ *Dissolved Air Flotation (DAF) system proposed to remove free Oil & Grease from the Effluent water*
- ✓ *To maintain optimum pH and Temperature in the Biological Oxidation System, Neutralization system and Cooling Tower proposed*
- ✓ *De-nitrification system proposed for complete conversion of ammoniacal Nitrogen into nitrogen gas*
- ✓ *Biological Oxidation system proposed with optimum retention time to maximize the pollutant reduction*
- ✓ *Lamella Settler proposed for efficient separation of solids and to provide polished influent to filtration systems*
- ✓ *Submerged Ceramic MBR system proposed to reduce Fouling Potential of RO feed water*
- ✓ *Contaminants removal system proposed to remove slowly Bio-degradable compounds present in the Effluent water* PATENTED
- ✓ *Color removal system proposed to remove the slight color of Biologically treated effluent water & to increase DO levels*
- ✓ *Degasser tower proposed to reduce alkalinity/bicarbonate by stripping CO_2 from effluent water*
- ✓ *Reverse Osmosis System proposed to achieve recovery of **95% in 4 Stages** with **Single Skid***

The following sections summarize the treatment processes, the major system components, the main system design parameters and design outputs as well as design details of system components. This section does not define scope of supply. Scope delineation is covered under section 3.

1.1. SYSTEM/TANK MODIFICATION DETAILS

EXISTING SYSTEM/TANK	PROPOSED SYSTEM	EXISTING SYSTEM USAGE	NEWLY PROPOSED SYSTEM
--	Rotary Brush screener	--	Completely new system proposed
Lifting Station	Lifting Sump	Existing Tank to be utilized after necessary modification	New Lifting Pump with piping & valves proposed
--	Dissolved Air Floatation (DAF)	--	Completely new system proposed with DAF Scraper, Bubble Generation pump & Sludge pump
Homogenizing Tank	Equalization tank	Existing Tank & Blower to be utilized as such	Diffusers with its piping & grid newly proposed
--	Neutralization System	Equalization to be utilized after partitioning for Biological Feed Pumps	Biological Feed Pump & Neutralization Dosing pumps with its piping & valves newly proposed
--	Cooling Tower	Existing Cooling tower of capacity 200m^3/hr utilized after necessary piping modifications	New Cooling tower proposed for expansion capacity of 100m^3/hr
Selector Tank	Denitrification Tank	Existing Tank to be utilized after necessary modification	Denitrification Tank flow mixers, Recirculation Pumps with its piping & valves newly proposed and the Existing Tank to be expanded / newly built

Oxidation tank	Biological oxidation tank	Existing Tank & Blowers to be utilized as such	Diffusers with its piping & Diffuser grid (PP-GF) newly proposed
Sedimentation Tank	Lamella Settler	Tank to be utilized for Lamella Settler after necessary modifications and the settler to be sufficient with a retention time of 4 to 6 hours.	Lamella Settler for expansion capacity of 100 M^3/Hr newly proposed & new RCC Tank to be built for expansion capacity
--	SRS system	Existing Tank to be utilized after necessary modification	New Sludge Recirculation pump with its piping & valves newly proposed
--	Submerged Ceramic MBR system	--	Completely new system proposed
--	Contaminant Removal System	--	Completely new system proposed
--	Color Removal System	--	Completely new system proposed
--	Degasser Tower	--	Completely new system proposed
--	Reverse Osmosis system	--	Completely new system proposed

1.2. INFLUENT QUALITY

S.NO	PARAMETER	UNIT	ETP INLET
1.	pH	--	9 –11
2.	BOD	mg/L	400
3.	COD	mg/L	1000
4.	TDS	mg/L	2500
5.	TSS	mg/L	< 250
6.	REACTIVE SILICA *	mg/L	< 15
7.	TOTAL NITROGEN	mg/L	30
8.	IRON *	mg/L	NIL
9.	TOTAL PHOSPHORUS	mg/L	< 2
10.	SULPHATE *	mg/L	< 800
11.	CHLORIDE *	mg/L	< 300
12.	HARDNESS *	mg/L of $CaCO_3$	< 120
13.	TOTAL ALKALINITY *	mg/L as $CaCO_3$	< 800
14.	TEMPERATURE *	°C	< 45
15.	OIL & GREASE	mg/L	< 20
16.	OTHER HEAVY METALS *	mg/L	NIL

IMPORTANT

NOTE: *Since the *Parametrical values are not available, it has been considered for the purpose of designing the treatment system. Any change in these values will impact design & cost of the ETP.*

TEMPERATURE: The Temperature has been provided as < 50 °C which may be a peak temperature. As per our service records, the average Temperature in similar plants would be < 45 °C. So, we have proposed separate cooling tower systems optionally to reduce the temperature from 50 °C to 40 – 45 °C.

The design solution proposed is based on the values as presented in the table above at ETP inlet. All concentrations refer to max concentrations to be used for the system's design. **Any change in the actual inlet parameters will have impact on Process Design, Engineering Design, Cost and Performance.**

1.3. INFLUENT FLOW DATA

PARAMETER (Feed Flow Rate)	VALUE	UNIT
ETP	7200	M^3/DAY
Degasser Tower & RO System	2400	M^3/DAY

1.4. OPERATIONAL BASIS

Hours Per Day of Operation	24
Days Per Year of Operation	350 days (Downtime for chemical cleaning and other regular annual maintenance work 15 days considered)

1.5. PROCESS COMPATIBILITY

1. The incoming raw wastewater shall not contain any substance incompatible with the MOC of the Equipment supplied & inhibitory substances that sufficiently affect the biological treatment stage so as to compromise the operation of any of the system.

2. The biological treatment is operated and maintained in accordance with good industry practice for wastewater treatment systems.

3. The Rotary Brush Screener at the plant inlet are installed and operated in such a way that fibrous material and floatable particles are removed reliably and any bypasses of the screens are impossible.

2. Design System Chosen

S.NO	PROCESS	CAPACITY (M^3/DAY)
1.	ROTARY BRUSH SCREENER	7200
2.	LIFTING PUMP	7200
3.	DISSOLVED AIR FLOTATION (DAF) SYSTEM	7200
4.	EQUALIZATION SYSTEM *(EXISTING BLOWER UTILIZED)*	7200
5.	NEUTRALIZATION SYSTEM	7200
6.	COOLING TOWER-2 *(FOR EXPANSION CAPACITY)*	2400
7.	DENITRIFICATION SYSTEM	7200
8.	BIOLOGICAL OXIDATION SYSTEM *(EXISTING BLOWER UTILIZED)*	7200
9.	LAMELLA SETTLER *(FOR EXPANSION CAPACITY)*	2400
10.	SLUDGE RECIRCULATION SYSTEM	7200
11.	SUBMERGED CERAMIC MBR SYSTEM	7200
12.	CONTAMINANTS REMOVAL SYSTEM PATENTED	720
13.	COLOUR REMOVAL SYSTEM	4800
14.	DEGASSER TOWER SYSTEM	2400
15.	REVERSE OSMOSIS SYSTEM **– (95% RECOVERY IN 4 STAGES)**	2400

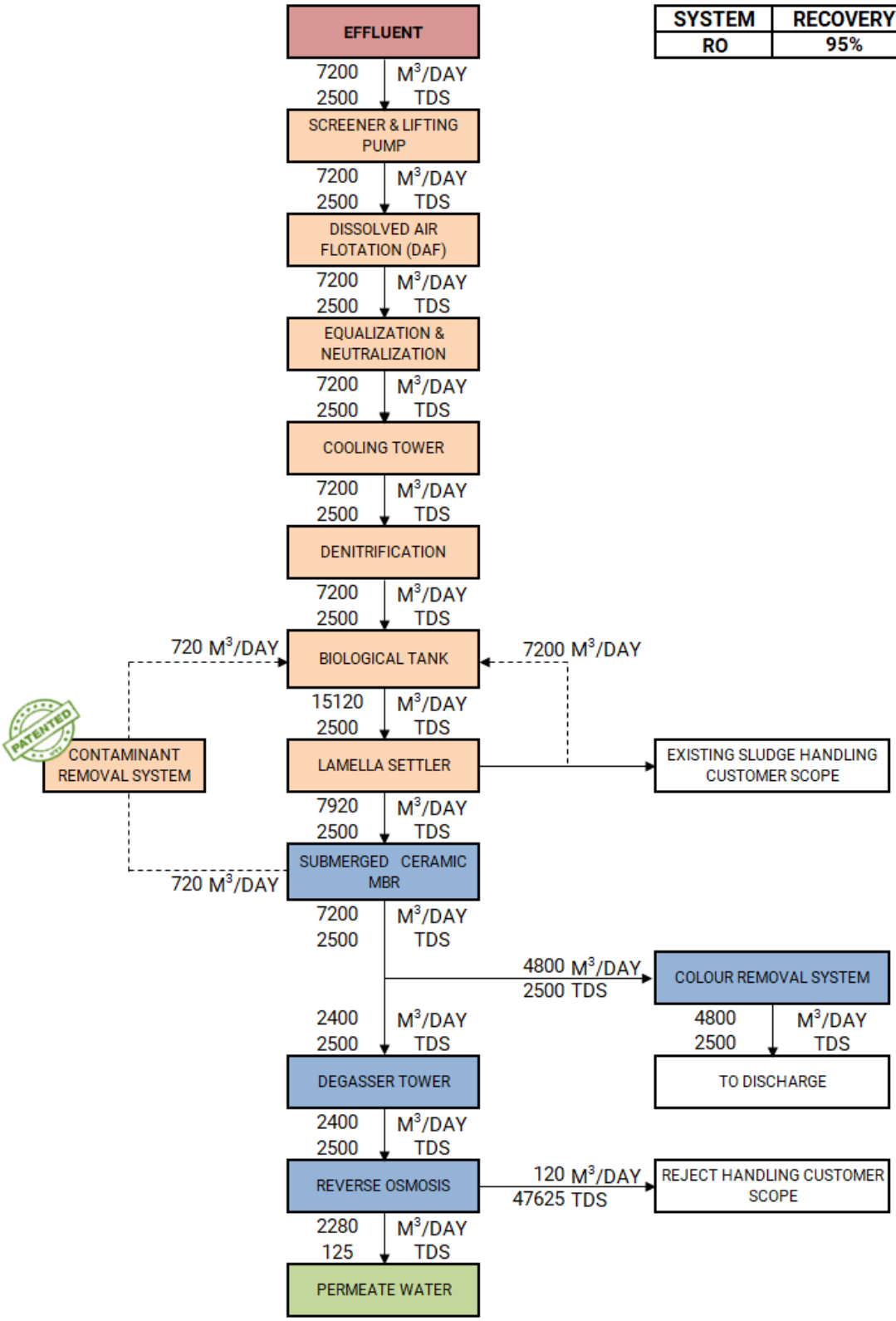
***System capacities mentioned above are based on ETP influent flow Rate and the Actual capacities may vary.**

Water Loving Technology

PROPOSED WATER BALANCE

ZLD BASED EFFLUENT TREATMENT FOR 7200 M³/DAY @ 2500 TDS

SYSTEM	RECOVERY
RO	95%



ETP - STAGEWISE PARAMETERS					
PARAMETER	UNIT	ETP INLET	COLOUR REMOVAL SYSTEM OUTLET	RO @ 95% RECOVERY	
				PERMEATE	REJECT
pH	--	9 –11	6 – 7	6 – 7	6 – 7
BOD	mg/L	400	< 30 *	BDL*	< 800 *
COD	mg/L	1000	< 80 *	BDL*	< 2000 *
TDS	mg/L	2500	Same as Inlet	< 150	47500 – 48500
TSS	mg/L	< 250	< 5	BDL	NA
REACTIVE SILICA	mg/L	< 15	Same as Inlet	< 1	< 320
TOTAL NITROGEN	mg/L	30	< 2	BDL	< 50
IRON	mg/L	NIL	NIL	NIL	NIL
TOTAL PHOSPHORUS	mg/L	< 2	Same as Inlet	BDL	NA
SULPHATE	mg/L	< 800	Same as Inlet	< 60	15000 – 17000
CHLORIDE	mg/L	< 300	Same as Inlet	< 50	5000 – 7000
HARDNESS	mg/L of $CaCO_3$	< 120	Same as Inlet	< 1	2500 – 3000
TOTAL ALKALINITY	mg/L as $CaCO_3$	< 800	Same as Inlet	< 20	< 2500
TEMPERATURE	ºC	< 45	30 – 35	30 – 35	30 – 35
OIL & GREASE	mg/L	< 20	< 1	BDL	NA
OTHER HEAVY METALS	mg/L	NIL	NIL	NIL	NIL

BDL – below detectable limit, NA – not applicable; Values are system generated which may vary with variation in inlet & on operational basis

* Depends on the Biological Degradability of the Effluent, considering **32 Hours of Retention time for Biological Treatment** and **available form of solubility. Also, it depends on the operational MLSS of the Biological Tank.**

NOTE: *Considering retention time of lamella settler as 4 - 6 hours for the existing ETP flow and the Lamella settler for Expansion capacity of 100 M^3/Hr has been newly proposed.*

2.1. EXPECTED TREATED WATER QUALITY

The following performance parameters are expected under standard operating conditions after equipment start-up based on the data and assumptions listed above and in Appendix, Warranties. In case of conflicting numbers, the ones listed under Appendix take precedence:

PARAMETER	TREATED WATER QUALITY	UNIT
TSS @ MBR PERMEATE	< 5	mg/L
DO @ COLOUR REMOVAL SYSTEM OUTLET	2 – 5	mg/L
TDS @ RO PERMEATE WITH 95% RECOVERY	75 – 150	mg/L

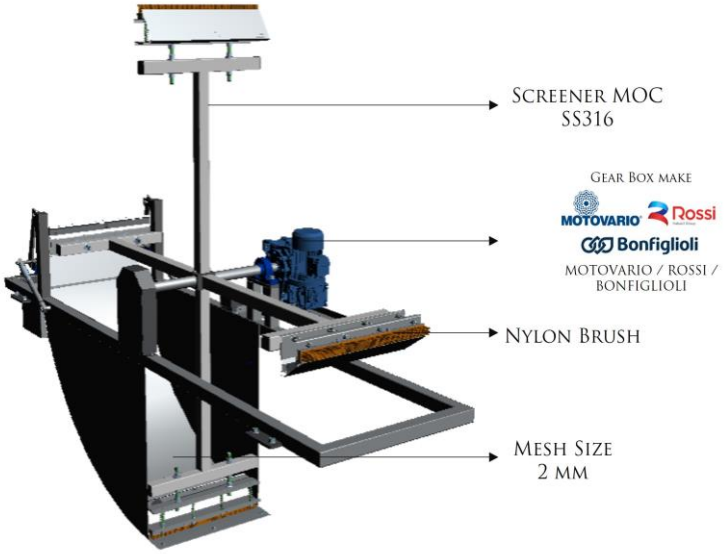
Notes:

- All influent quality parameters are based on daily average values of a minimum of four (4) Nos. of hourly composite samples collected at regular intervals over a day, with testing performed to applicable industry-approved standards.
- The treated water parameters could take 6 months for attaining specified values as per above table after successful commissioning.
- The above recovery water quality expectation is based on WTT supplying the system and scope Equipment as per the scope of supply table described in section 3 below.

3. Equipment Details for Proposed Systems

3.1. Rotary Brush Screener

Automated Brush Screener separates coarse & medium fine solids of size above 2 mm from influent. This process is a predetermined stage where escaping of solids is completely avoided, where by clogging of pumps & machinery in subsequent steps gets nullified. The collected wastes have to be disposed periodically and the screener is attached with brush.



3.2. Dissolved Air Flotation (DAF) System

DAF system used to remove total suspended solids (TSS) and oils & greases (O&G) from effluent water; this effectively reduces the pollutant load in MBR system. Air and water suctioning pump provided for the flotation of suspends and oils. Scrapping mechanism provided for highly efficient separation of floating suspends and Oils & Greases.

3.3. EQUALIZATION SYSTEM

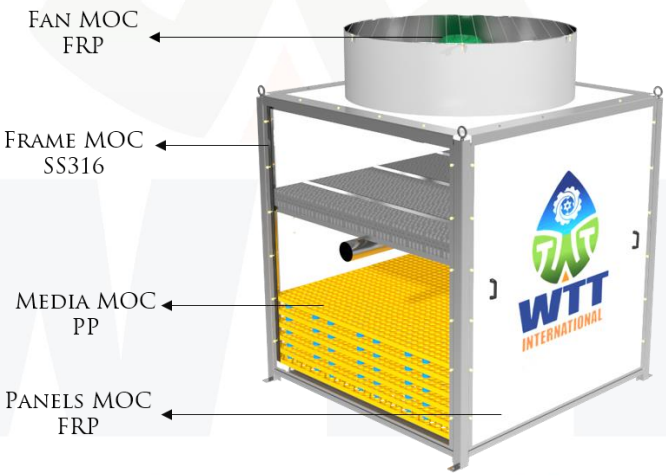
Equalization system used to homogenize different wastewater characteristics to achieve uniform pollutant load. The continuous mixing and water movement maintained by diffused aeration to avoid the dead zone, solids sedimentation and anaerobic fermentation.

3.4. NEUTRALIZATION SYSTEM

The Biological oxidation system requires a neutral or slightly alkaline pH value for the optimum performance. Neutralization is carried out automatically depending on the pH of the inlet water.

3.5. COOLING TOWER

Cooling tower is a specialized heat exchanger in which air and water are brought into direct contact with each other in cross flow to reduce the inlet water temperature and it is designed to handle high TSS effluent. It is recommended to keep and maintain the biological oxidation system in optimum temperature for better performance.



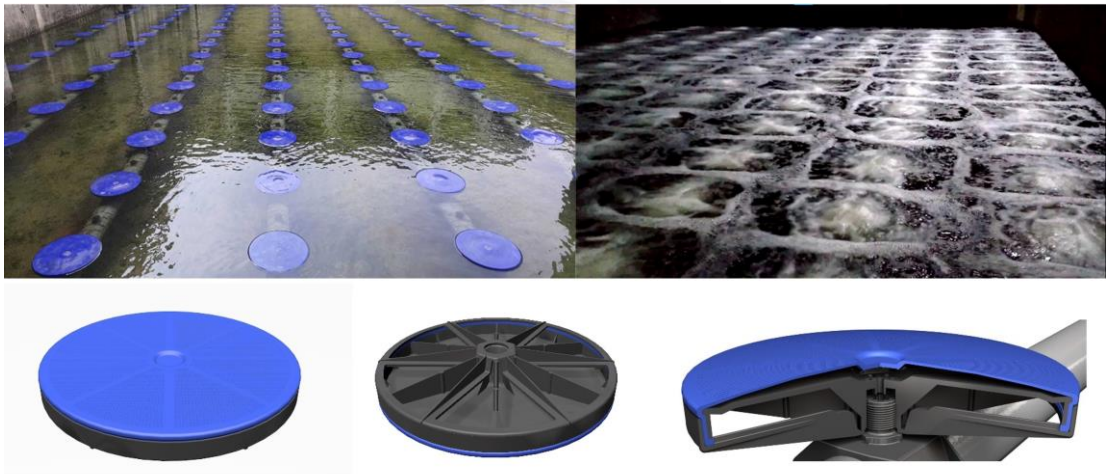
3.6. DENITRIFICATION SYSTEM

Anoxic digestion is a series of biological processes in which microorganisms break down the biodegradable material in the absence of oxygen. Nitrate nitrogen converted into nitrogen gas in the absence of dissolved oxygen.

NO_3^- →	NO_2^- →	NO →	N_2O →	N_2
Nitrate	Nitrite	Nitric Oxide	Nitrous Oxide	Nitrogen Gas

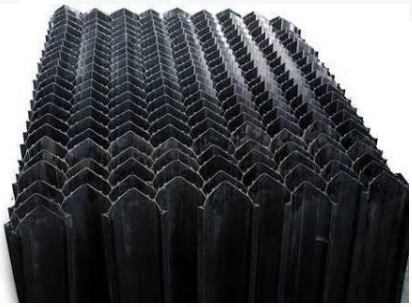
3.7. BIOLOGICAL OXIDATION SYSTEM

The organic matter is aerated in biological tank to bring down the pollutant load using micro-organisms. The micro-organisms metabolize the organic matters and a Part of organic matter is synthesized into new cells. The microbial growth in the wastewater maintaining by providing required DO level.



3.8. LAMELLA SETTLER

Lamella settler system removes solid particulates and suspend solids from liquid through gravity settling to provide polished influent for filtration process and also ensures the required retention time for settling of suspends.



3.9. SUBMERGED CERAMIC MBR SYSTEM

This system comprises of Eco-friendly & long-lasting Ceramic Membranes. Ceramic MBR system operates at low operational flux and easy to clean as the membrane is rigid and do not lose its form over a period of time. The system can be cleaned with Permeate & Chemically Enhanced Backwash technology and Air scouring to avoid sludge deposition. The sprinkler pumps ensures that there is no clogging over the membrane surface.

3.10. CONTAMINANT REMOVAL SYSTEM

Contaminant Removal system reduces the circulation of inorganic compounds in biological oxidation system. These compounds can be removed from the membrane surface by backwashing and the drained water has to be treated before being fed back to the biological oxidation system. Where it does not settle at the biological process and the repeated circulation of these compounds in biological feed stream will decrease the efficiency of biological oxidation system so this inorganic compound will completely remove via Contaminant Removal System

3.11. COLOUR REMOVAL SYSTEM

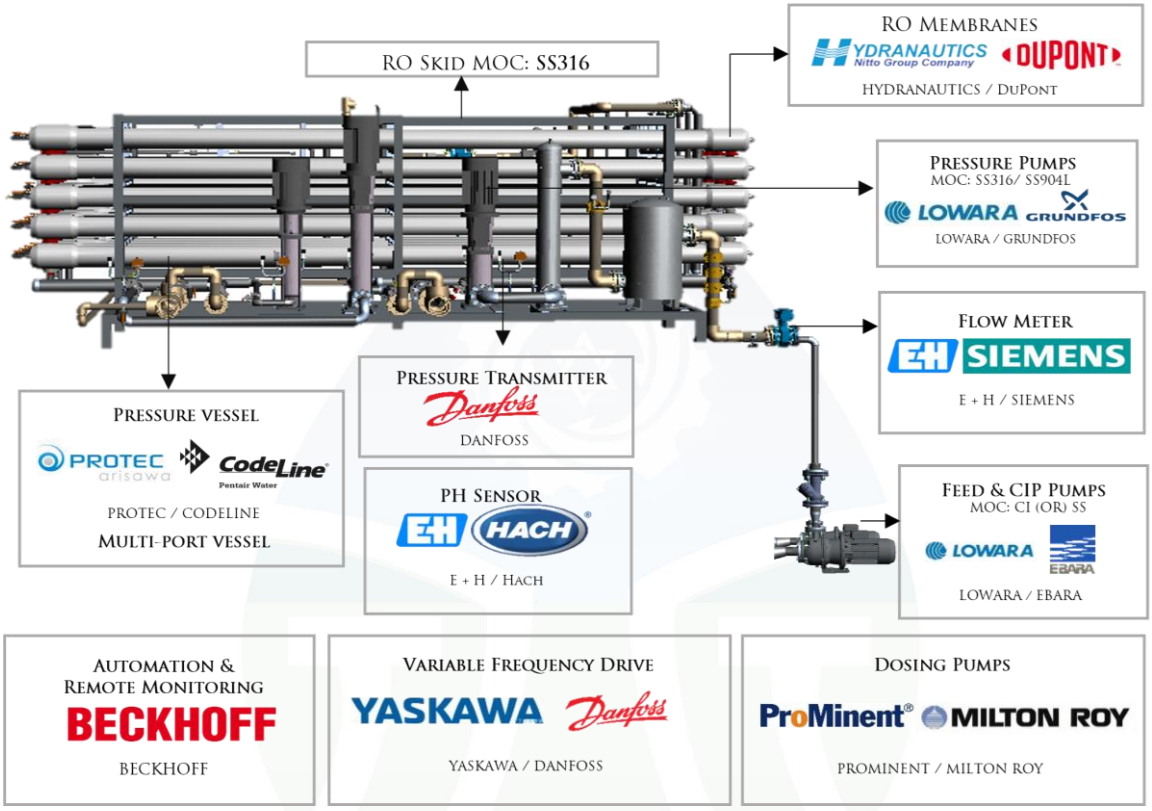
Color removal system is proposed to reduce color and COD of the effluent. The clarified effluent will have slight yellowish color and traces of COD. Hence it has to be chlorinated. The chlorination not only removes color but also disinfects the water to be discharged. Also, it increases the DO level of the Discharge water.

3.12. DEGASSER TOWER SYSTEM

Degasser tower eliminates the carbonaceous gases from the effluent water by means of a packed decarbonizing tower. The air blown from the bottom of the tower carries away CO_2 from wastewater to the top of the tower to reduce the dissolved gases and there by protect the RO membranes from scaling

3.13. REVERSE OSMOSIS SYSTEM

Reverse Osmosis Process is the forced passage of water through a membrane against the natural osmotic pressure to accomplish separation of ions and water. It separates the dissolved salts from the feed water and reduces the TDS level, and this can be reused for industrial process. We can achieve lowest CAPEX and OPEX.



4. Equipment Scope of Supply – BY WTT

PLANT EQUIPMENT LIST			
A	**BIOLOGICAL SYSTEM**		**7200 M^3/DAY**
S.NO	**DESCRIPTION**	**SPECIFICATION**	**QUANTITY**
1.	SCREENER & LIFTING PUMP	ROTARY BRUSH SCREENER	1 No.
		MESH SIZE	2 MM
		MOC	SS316
		LIFTING PUMP	3W + 1S
		PUMP MOC (CASING/IMPELLER)	CI / CI
		PUMP TYPE	SUBMERSIBLE
2.	DISSOLVED AIR FLOTATION SYSTEM	DAF SCRAPER MECHANISM	3
		SCRAPER MECHANISM MOC	SS316

		DAF CIRCULATION PUMP	3W + 1S
		PUMP TYPE	AIR-WATER SUCTIONING TYPE (**NANO BUBBLE GENERATION PUMP**)
		PUMP MOC (CASING/IMPELLER)	CI / SS
		DAF SLUDGE PUMP	1W + 1S
		PUMP TYPE	SURFACE MOUNTED
		PUMP MOC (CASING/IMPELLER)	CI / CI
3.	EQUALIZATION TANK	DIFFUSER	296 No's
		DIFFUSER SIZE	11"
		DIFFUSER TYPE	FINE BUBBLE
		DIFFUSER MOC	PP-GF disc with SILICON membrane
		DIFFUSER GRID PIPING MOC	SS316L
4.	BIOLOGICAL FEED SYSTEM	BIOLOGICAL FEED PUMPS	3W + 1S
		PUMP MOC (CASING/IMPELLER)	CI / CI
		PUMP TYPE	SUBMERSIBLE
		pH SENSOR	1
		pH SENSOR RANGE	0 – 14
		NEUTRALIZATION DOSING PUMPS	1W + 1S
		PUMP MOC	TEFLON
		ELECTROMAGNETIC FLOWMETER	1
		COOLING TOWER **(DOUBLE BAY)**	1 *
		COOLING TOWER FAN	2 NOS.
		COOLING TOWER MEDIA	2 LOT
		MEDIA MOC	PP

5.	DENITRIFICATION SYSTEM	DENITRIFICATION PUMP	4W + 1S
		PUMP MOC (CASING/IMPELLER)	CI/CI
		PUMP TYPE	SUBMERSIBLE
		FLOW MIXER FOR DENITRIFICATION SYSTEM	1 LOT
		FLOW MIXER PROPELLER MOC	SS
6.	BIOLOGICAL SYSTEM	DO SENSOR	2
		DO SENSOR RANGE	0 – 5 PPM
		BLOWER	1 STANDBY
		BLOWER TYPE	LOBE
		DIFFUSER	2519 No's
		DIFFUSER SIZE	11"
		DIFFUSER TYPE	FINE BUBBLE
		DIFFUSER MOC	PP-GF disc with SILICON membrane
		DIFFUSER GRID PIPING MOC	SS316L
7.	LAMELLA SETTLER & SLUDGE RECIRCULATION SYSTEM	LAMELLA SETTLER	1 LOT *
		LAMELLA SETTLER FRAME MOC	SS316
		LAMELLA PACKS MOC	PVC
		SLUDGE RECIRCULATION PUMP	3W + 1S
		PUMP MOC (CASING/IMPELLER)	CI / CI
		PUMP TYPE	SURFACE MOUNTED

*** Equipment has been proposed only for Expansion capacity of 100 M^3/Hr.**

Existing Equalization & Biological Blowers with inverter, cooling tower to be utilized as such.

B	SUBMERGED CERAMIC MBR SYSTEM		7200 M^3/DAY
S.NO	DESCRIPTION	SPECIFICATION	QUANTITY
8.	SUBMERGED CERAMIC MBR SYSTEM	NO. OF MODULES	3 LOT
		NO. OF TRAINS	3
		MOC OF MODULE	CERAMIC
		PERMEATE PUMP	3W + 1SS
		PUMP MOC (CASING/IMPELLER)	CI / CI
		TYPE OF PUMP	SURFACE MOUNTED
		BACKWASH/CIP PUMP	1W + 1S
		PUMP MOC (CASING/IMPELLER)	CI / CI
		TYPE OF PUMP	SURFACE MOUNTED
		SLUDGE PUMP	1W + 1S
		PUMP TYPE	SURFACE MOUNTED
		PUMP MOC (CASING/IMPELLER)	CI / CI
		DOSING PUMP	2W + 2S
		DOSING PUMP MOC	PP
		BLOWER	1W + 1S
		BLOWER TYPE	LOBE TYPE WITH ACOUSTIC HOOD
		TURBIDITY SENSOR	PERMEATE
		ELECTROMAGNETIC FLOW METER	PERMEATE

C	CONTAMINANTS REMOVAL SYSTEM		720 M^3/DAY
S.NO	DESCRIPTION	SPECIFICATION	QUANTITY
9.	CONTAMINANTS REMOVAL	LAMELLA SETTLER	1 LOT

	SYSTEM	LAMELLA PACKS MOC	PVC
		ANGLE OF INCLINATION FR PACKS	55 – 60°
		LAMELLA SETTLER FRAME MOC	SS316
		DOSING PUMP	1W + 1S
		DOSING PUMP MOC	PP
		SLUDGE PUMP	1W + 1S
		PUMP MOC (CASING/IMPELLER)	CI/CI
		PUMP TYPE	SURFACE MOUNTED

D	COLOR REMOVAL SYSTEM		4800 M^3/DAY
S.NO	**DESCRIPTION**	**SPECIFICATION**	**QUANTITY**
10.	COLOR REMOVAL SYSTEM	DOSING PUMP	1W + 1S
		DOSING PUMP MOC	PP
		CIRCULATION PUMP	1W + 1S
		PUMP TYPE	SURFACE MOUNTED

E	DEGASSER TOWER		2400 M^3/DAY
S.NO	**DESCRIPTION**	**SPECIFICATION**	**QUANTITY**
11.	DEGASSER TOWER	DEGASSER TOWER VESSEL	1
		TOWER MOC	FRP
		pH SENSOR	1
		DEGASSER TOWER MEDIA	1 LOT
		MEDIA MOC	PP
		DEGASSER TOWER BLOWER FAN	1
		DOSING PUMP FOR DEGASSER TOWER	1W + 1S

		PUMPS MOC	PP / TEFLON

F	REVERSE OSMOSIS SYSTEM		2400 M^3/DAY
S.NO	**DESCRIPTION**	**QUANTITY**	**QUANTITY**
12.	REVERSE OSMOSIS SKID	**TOTAL NUMBER OF SKIDS**	**1**
		SKID MOC	SS316
		NUMBER OF STAGES	4
		RO FEED PUMP	1W + 1S
		PUMP MOC (CASING/IMPELLER)	CI / CI
		PUMP TYPE	SURFACE MOUNTED
		PRE-FILTER	1 LOT
		PRE-FILTER HOUSING MOC	SS316
		DOSING PUMPS	3W + 1S
		DOSING PUMPS MOC	PP
		DOSING TANKS	2
		DOSING TANK MOC	LDPE
		HIGH PRESSURE PUMP WITH INVERTER	1W
		HIGH PRESSURE PUMP MOC	SS316
		HIGH PRESSURE PUMP TYPE	VERTICAL/ MULTI STAGE
		BOOSTER PUMPS WITH INVERTER	3W
		BOOSTER PUMPS MOC	SS316
		BOOSTER PUMP TYPE	VERTICAL/ MULTI STAGE
		PRESSURE VESSELS	31 No's
		PRESSURE VESSELS MOC	FRP
		RO MEMBRANES	186 No's
		RO MEMBRANES MOC	COMPOSITE POLYAMIDE

		pH SENSOR	1
		PRESSURE INDICATOR	10
		PRESSURE TRANSMITTER	10
		ELECTROMAGNETIC FLOWMETER	5 No's (RO FEED, EACH STAGE PERMEATE LINE)
		CIP DOUBLE SKID WITH AUTOMATION (INCLUSIVE OF PUMPS WITH ITS FILTER, pH SENSOR FLOWMETER, PRESSURE TRANSMITTER, AUTO VALVES & DOSING PUMPS)	1 SET

W – Working; S – Standby; SS – Store Standby

Specification and quantity details mentioned above are preliminary and may be revised to optimize the performance of the system.

5. OPTIONAL SYSTEM

5.1. SUBMERGED HOLLOW FIBER MBR SYSTEM

This system combines conventional biological treatment processes with membrane filtration to provide an advanced level of organic and suspended solids removal in reduced foot print. Both clarification and filtration provide polished influent for RO system.

A	SUBMERGED HOLLOW FIBER MBR SYSTEM		7200 M^3/DAY
S.NO	**DESCRIPTION**	**SPECIFICATION**	**QUANTITY**
1.	SUBMERGED HOLLOW FIBER MBR SYSTEM	MBR MODULE	9 NOS.
		FILTRATION AREA	16200 M^2
		MOC OF MODULE	PVDF, Braided hollow Fiber
		MOC OF RACK	SS316

		PERMEATE/BACKWASH/CIP PUMP	3W + 1SS
		PUMP MOC	CI
		TYPE OF PUMP	LOBE TYPE
		OPERATION TYPE	REVERSIBLE LOBE
		BLOWER	1W + 1S
		BLOWER TYPE	LOBE
		SLUDGE PUMP	1W + 1S
		PUMP MOC	CI / CI
		PUMP TYPE	SURFACE MOUNTED PUMP
		DOSING PUMP	3W + 3S
		DOSING PUMP MOC	PP
		ELECTROMAGNETIC FLOW METER	PERMEATE
		ELECTROMAGNETIC FLOW METER MAKE	E+H / SIEMENS

W – Working; S – Standby; SS – Store Standby

Specification and quantity details mentioned above are preliminary and may be revised to optimize the performance of the system.

OTHER GENERAL ITEMS IN WTT's SCOPE:

- *Piping in SS, CPVC, UPVC and its associated items*
- *All other related valves and puddle flanges required*
- *Electrical control panel for proposed system with standard automation*
- *Power cable and control cable within the electrical panel*
- *Power cable and control cable from electrical panel to proposed equipment*
- *General layout & civil detailed drawing for proposed system*

6. Scope Of Supply & Installation- By Client

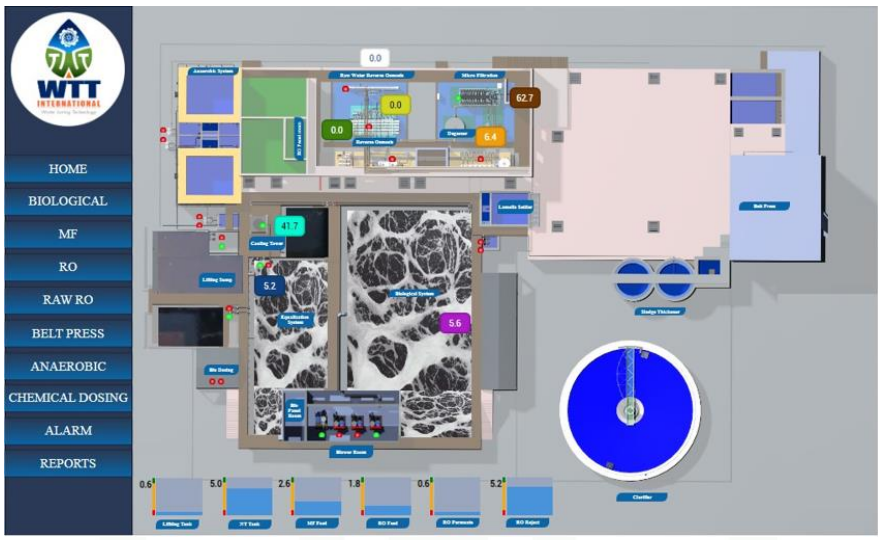
- Installation of supplied Equipment
- Transfer of Effluent up to proposed Screener system
- Existing Equalization & Biological oxidation system Blower with inverter
- Existing Cooling Towers to be utilized as such
- Existing Lamella settler of capacity 200 M^3/Hr with at least 4 to 6 Hours of Retention time
- Aeration / Frequency control for Equalization Blower as the Air required for Equalization system is about 600 M^3/Hr
- Sludge Handling of Proposed ETP Plant
- Discharge of Treated water
- RO Permeate water transfer for Reuse
- Bulk chemical storage tanks for proposed systems based on the available optimum level of chemical procurement
- Sludge disposal during plant commissioning and operation
- Backwash/CIP Tank (RCC) for Proposed MBR system
- Raw / Permeate water transfer for cleaning of filtration Plant
- Transfer of CIP cleaning water from Filtration Plant
- Raw water supply for whole plant
- Air conditioning Systems for Panel room & Personal Computers
- RO Reject Water Transfer and handling
- All underground piping & piping for electrical cable during civil works
- Incomer cable and its connections with WTT supplied electrical panel
- Two Separate & Distinct Earthing for all individual equipment & panel
- Instruments/Electronics Earthing & Safety/Protective Earthing shall be provided for individual Instruments, Motor/Pump & Other Electrical Equipment
- Input power to individual WTT supplied electrical panel
- All civil related works, civil structural drawings, underground piping & piping for electrical cabling during civil works
- Installation of equipment supplied
- All loading & unloading @ site

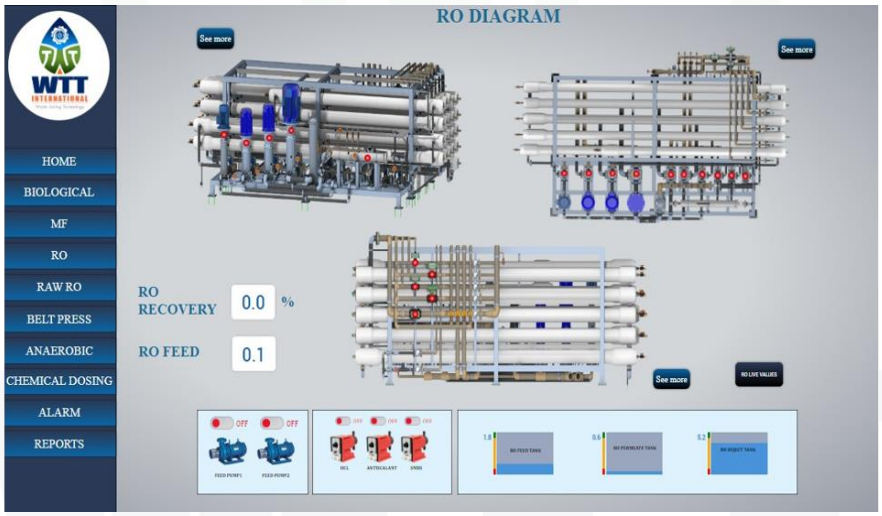
- All excavation works related with pipe and cables laying
- Electrical cable & stabilized power supply until WTT panel's, Internal Individual Earthing as per requirement
- Utilities like Power, Steam, Hot water etc., if Required
- Chemical supply and storage/preparation tanks
- Separate panel, wiring with fans & lights for whole plant & its power supply
- Pre-Shipment Inspection requested by the local authorities
- All license fees and / or custom duties
- Provide Food, Accommodation and Local transport to the team of WTT
- Installation consumables (such as welding electrodes, cutting disk and others)
- Supply and assembly of all drainage lines for rain water, sewers, etc
- Any construction approval from local Authorities
- Supply of any hoisting and/or laying equipment (cranes forklift, etc.) both truck mounted or not
- Construction of site fencing/railing
- Protection for all field instruments, local switches, motors, etc.,
- Safety handrails for all tanks, safety items like life buoy rings, eye washer, life-jackets, etc
- Removal of all residual debris after construction and thorough cleaning of all tanks before plant start-up
- General spares for supplied equipment
- Any further item not described in our offer

HIGHLIGHTS

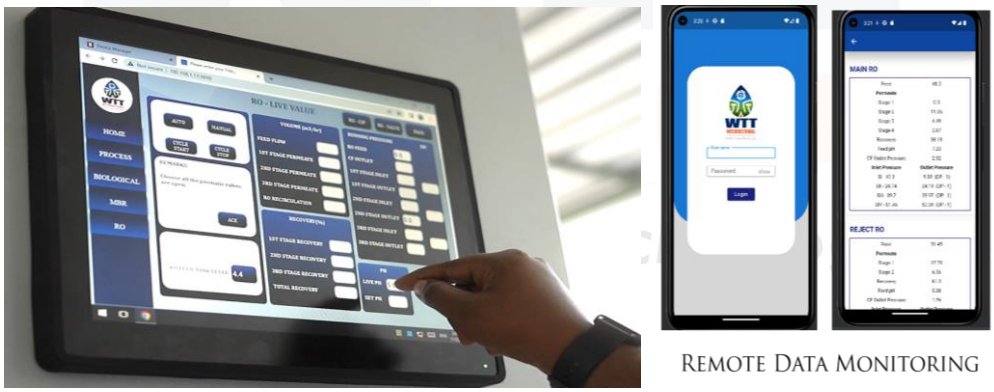
- *Our scope of supplies is of **SS316L pipes for Air line, MBR pipe line, RO pipe line**, materials like UPVC are used only in areas where SS316L is not compatible.*
- *Rotary Brush Screener fabricated at a world class manufacturing facility with SS316 having sturdy axial rotating brush arm with long-lasting **Nylon Bristles** enables screening by employing punched hole sheets up to **mesh size of 2mm**.*
- *We are supplying the DAF system with **NIKUNI - Japan pumps** for **Nano bubble generation** which is Key Importance for our process.*
- ***DO Sensor** based Automation for Biological Aeration to maintain balanced F/M ratio.*
- *Diffusers made of long-lasting **PP-GF Silicon membrane** and enables utmost mixing by fine bubbles.*
- *Submerged CERAMIC MBR system with a **higher membrane life** and auto cleaning capabilities.*
- ***Rapid Filtration & Backwash Sequence** with automatic valves for pre-filtration.*
- *Specialized Degasser Tower proposed to **reduce the scaling of RO membranes** and in-turn achieves lesser number of cleaning cycles.*
- *This Degasser tower system enables to **increase the life of RO membranes.***
- *All part of the plant has **Level sensors and floats** to have complete data on volume of water in each tank.*
- ***Individual pressure transmitters for Inlet & Outlet** of each stage element to know exactly where the fouling has occurred and to initiate cleaning for that specific stage.*
- *Each stage of **RO permeate has individual EMFM** for better control and better automation to keep efficiency and life of RO.*
- ***STATE OF ART 2 SKIDS OF CIP system** for RO with full automation including dosing of chemicals with pH automation and flow with pressure control.*
- *Electrical panel supplied by WTT with necessary cooling technologies and easy operative switch for Automatic & Manual operations.*

- ***PC controlled HMI panel*** *provided with complete automation and acknowledgement & warnings for cleaning necessities based on the operating conditions.*





Sample of our HMI screens and is a tentative reference



- *Making the plant In-hand through* ***SCADA*** *for Live Data monitoring using a smartphone (or) PC.*

7. Typical Equipment Vendor List

S.NO	EQUIPMENT	VENDORS
1.	DIFFUSER	OTT / GEOTIERRE
2.	MBR BLOWER	ROBUSCHI
3.	SUBMERSIBLE PUMP	GRUNDFOS / XYLEM / EBARA
4.	PRESSURE PUMP	GRUNDFOS / LOWARA
5.	SURFACE MOUNTED PUMP	EBARA / LOWARA
6.	DOSING PUMP	PROMINENT / MILTON ROY
7.	CERAMIC MBR MEMBRANES	OVIVO
8.	HOLLOW FIBER MBR MEMBRANES	KOVALUS (KOCH)
9.	RO MEMBRANES	HYDRANAUTICS / DUPOINT
10.	PRESSURE VESSELS	CODELINE / PROTEC
11.	VARIABLE FREQUENCY DRIVE	YASKAWA / DANFOSS
12.	ELECTROMAGNETIC FLOWMETER	E+H / SIEMENS
13.	INLINE ANALYZERS	E+H / HACH
14.	LEVEL TRANSMITTER	E+H / SIEMENS
15.	LEVEL FLOAT	FAES
16.	PRESSURE TRANSMITTER	DANFOSS
17.	PRESSURE GAUGE	FORBES MARSHALL / MASS / WIKA
18.	AUTOMATION	BECKHOFF
19.	ELECTRICAL PANEL	RITTAL / HOFFMAN
20.	ELECTRICAL CABLE	LAPP / HUBER+SUHNER
21.	ROTARY BRUSH SCREENER	**WTT INTERNATIONAL**
22.	DISSOLVED AIR FLOTATION SYSTEM	
23.	COOLING TOWER	
24.	LAMELLA SETTLER	
25.	CONTAMINANTS REMOVAL SYSTEM	
26.	DEGASSER TOWER	
27.	RO SKID	

The above-mentioned vendors are subjected to change based on the delivery terms for an equivalent vendor without compromising the quality and WTT reserves the right to change the vendor without any prior intimation.

8. Pricing Detail

PRICE FOR PROPOSED SYSTEMS		
S.NO	**SYSTEM/EQUIPMENT**	**PRICE IN USD**
1.	**BIOLOGICAL SYSTEM – 7200 M³/DAY** (WHICH INCLUDES SCREENER, LIFTING PUMP, DISSOLVED AIR FLOTATION, EQUALIZATION SYSTEM, NEUTRALIZATION SYSTEM, COOLING TOWER, BIOLOGICAL SYSTEM, LAMELLA SETTLER, SRS)	**$ 1,471,202**
2.	**SUBMERGED CERAMIC MBR** WITH CONTAMINANT REMOVAL SYSTEM**– 7200 M³/DAY**	**$ 3,093,864**
3.	**COLOUR REMOVAL SYSTEM – 4800 M³/DAY**	**$ 36,315**
4.	**DEGASSER TOWER SYSTEM – 2400 M³/DAY**	**$ 48,029**
5.	**REVERSE OSMOSIS SYSTEM – 2400 M³/DAY** (**95% RECOVERY** IN 4 STAGES WITH **SINGLE SKID**)	**$ 748,236**
TOTAL PRICE FOR PROPOSED SYSTEMS		**$ 5,397,646**

- ***Cooling Tower & Lamella settler has been proposed only for Expansion capacity of 100 M³/Hr.***
- ***Contaminant removal system has been proposed suitable for the Proposed Submerged Ceramic MBR & Hollow Fiber MBR systems***

OPTIONAL PRICE FOR PROPOSED SYSTEMS		
S.NO	**SYSTEM/EQUIPMENT**	**PRICE IN USD**
1.	**RO HIGH PRESSURE PUMP** - STORE STANDBY FOR ALL 4 STAGES	**$ 44,047**

PRICE FOR OPTIONAL SYSTEMS		
S.NO	**SYSTEM/EQUIPMENT**	**PRICE IN USD**
1.	**COOLING TOWER SYSTEM – 7200 M³/DAY** **(IN CASE OF AVERAGE TEMPERATURE LESS THAN 50 °C)**	**$ 96,500**

2.	**SUBMERGED HOLLOW FIBER MBR SYSTEM – 7200 M^3/DAY**	**$ 2,744,532**

All above mentioned prices given for the proposed systems with complete automation

- Pricing provided herein does not include any duties and taxes
- Pricing provided herein is on **Ex-Works** basis & does not include the insurance charges
- Supervision of erection and commissioning of WTT supplied system shall be extra
- Payment/commercial terms & conditions as per WTT's General Terms & Conditions of sale.
- Price valid for 15 days from the date of proposal due to market volatility.

9. Happy Customers

WELSPUN
JCT LIMITED
SIDDHI Group
NSL TEXTILES
Sharadha TERRY PRODUCTS LTD
BHILWARA LNJ GROUP
KPR MILL LIMITED
Kanchan
TRIDENTGROUP
Vardhmän
BEST
KITEX
Sona
SKT
BIRLA CENTURY
Himatsingka
premier
GANGA Acrowools
NAHAR INDUSTRIAL ENTERPRISES LIMITED
rdl RSL Dyecot Private Lmited
NUMALIGARH REFINERY LIMITED
NRL
silk park kanchipuram
Ministry of Textile
Government of India & Government of Tamilnadu

AND 12 CETPs ACROSS THE COUNTRY

10. Channel Partners

HYDRANAUTICS Nitto Group Company
DUPONT
AsahiKASEI
KOCH MEMBRANE SYSTEMS
xylem Let's Solve Water
GRUNDFOS
ROBUSCHI
Atlas Copco
KAESER COMPRESSORS
Endress+Hauser People for Process Automation
HACH
ABB
SIEMENS
BECKHOFF
RITTAL
OTT GROUP

11. Process Description Video

12. Commercial Terms and Conditions

12.1. Taxes

Taxes and Duties shall be extra as applicable. Pricing provided herein does not include any taxes of destination country.

If Tax Exemption is applicable, customer has to provide a copy to WTT of any applicable tax exemption certificates as issued by an approved taxation authority for the specific project location. Without an approved tax exemption certificate received by WTT all submitted invoices will include applicable tax. However we reserve the right to revise our price in case of tax exemption.

12.2. Freight

All the Equipment and components foresee **Ex-works** delivery as per I.C.C. Incoterms-2010.

12.3. Invoicing and Payment Terms

The pricing quoted in this proposal is based on the following payment terms, in principally agreed.

(1) 30% Advance Payment along with PO.

(2) 70% by irrevocable Letter of Credit payable at sight against Shipment documents.

Equipment shipment is contingent on receipt of earlier milestone payments.

12.4. Equipment Shipment

Client and WTT will arrange a kick off meeting after contract acceptance to develop firm shipment schedule. This estimated delivery schedule assumes no more than 1 week for Client review of submittal drawings. Any delays in Client approvals or requested changes may result in additional charges and/or a delay to the schedule.

Material supply for the proposed plant in 6 - 7 months (approx.) is expected. The delivery schedule excludes the months of August & December, the delivery schedule is subject to review and adjustment. Partial shipments are allowed. In case of modifications to the volume of the supply after the emission of the order confirmation, WTT reserves the right to modify the delivery time and the agreed price.

12.5. FORCE MAJEURE

Please be advised that force majeure is applicable to the quotation provided. The unforeseen and uncontrollable events as defined in our terms and conditions, such as natural disasters, wars or pandemics & other events that prevent the parties from fulfilling their contractual commitments have occurred, impacting our ability to honor the terms outlined in this quotation. We will diligently review the situation and promptly communicate any necessary adjustments or potential delays in fulfilling the quoted services or products in compliance with the force majeure provisions outlined in our agreement.

12.6. PRICING NOTES

All prices are quoted in **USD**. Any applicable tax of destination country is not included. Client will pay all applicable Local, State / Provincial, or Federal taxes & Duties. WTT may manufacture and source the Equipment and any part thereof globally in the country or countries of its choosing, provided that the Equipment complies with all of the requirements specified in this Agreement.

The Equipment delivery date, start date, and date of commencement of operations are to be negotiated. Title and risk of loss will transfer upon delivery in accordance with the INCOTERMS 2010. Commercial Terms and Conditions shall be in accordance with WTT's General Terms and Conditions of Sale as included in Appendix.

13. CLIENT SCOPE OF SUPPLY

13.1. SAFETY AND ENVIRONMENTAL

First aid, emergency medical response, eyewash & safety showers in the water treatment area. Chemical spill response, security & fire protection systems as per local codes. Environmental use and discharge permits for all chemicals at Client facility either listed in this document or proposed for use at a later date. Any special permits required for WTT's or Client employees to perform work related to the water treatment system at the facility. All site testing, including soil, ground and surface water, and air emissions, etc. Disposal of all solid & liquid waste from WTT's system including waste materials generated during construction, startup and operation.

Provide appropriate protection of the environment & local community, the health and safety of all workers and visitors at the site and the security of the facility. Provide safety related equipment & services such as site security, fire systems, lifting equipment and its operation, fall protection, adequate floor grating, ventilation, and safe access to equipment & electrical systems areas.

Equipment and trained support personnel for any confined space entry required during equipment installation/startup/commissioning/servicing. For permit-required confined space entry, a qualified rescue team on stand-by and available to respond within 4 minutes of an emergency.

Client will identify and inform WTT's personnel of any hazards present in the work place that could impact the delivery of WTT's scope of supply and agrees to work with WTT to remove, monitor, and control the hazards to a practical level. Client will provide training to WTT's personnel on all relevant & standard company operating procedures and practices for performing work on site.

13.2. JOBSITE AND INSTALLATION REVIEW

Review of WTT's supplied equipment drawings and specifications. All easements, licenses and permits required by governmental or regulatory authorities in Connection with the supply, erection and operation of the system.

Overall plant design, detail drawings of all termination points where WTT's equipment or Materials tie into equipment or materials supplied by others Stamping, signing or Sealing General drawings as per State, or local regulations or codes. All applicable civil design and works, including any building, site preparation, grading, excavations, foundations and trenches and accessories.

All electrical Labor and supplies leading up to jobsite including fittings, conduit, supports, cable trays, wire and hardware, air-conditioning of panels required to appropriately ground / earth the equipment as required for installation and ongoing operations.

All mechanical Labor and supplies leading up to jobsite including interconnecting piping, heat tracing (if required), fittings, conduit, pipe supports, and hardware as required for installation and ongoing operations. All instrumentation and automatic pneumatic valves

including but not limited to; air/sample line tubing, fittings, conduit, supports, isolating valves as required for installation and ongoing operations. Loading, unloading and transportation of equipment, materials required for WTT to perform duties outlined in WTT's Scope of Supply to the jobsite and/or warehouse.

Client will provide all access structures (scaffolding), mechanical lifting equipment (cranes, forklifts, scissor lifts, etc.), suitable site/shelter for placement of the proposed equipment, either inside appropriate housing, or outdoors.

Precaution: electrical equipment including the PLC may require air-conditioned rooms to prevent overheating of sensitive electronic equipment depending on climatic conditions. Bulk chemical storage and tanks, including secondary containment in accordance with local codes. Client will receive, off-load, log, and store all chemical and materials in accordance with Manufacturer's recommendation that are shipped. Compressed instrument air for pneumatic valves instruments and ejector system Equipment anchor bolts, Laboratory services, operating and maintenance personnel during equipment checkout, start-up and operation. Any on-site painting or touch-up painting of equipment supplied. Disposal of any Preservative.

For erection, WTT will ship the accessories such as pipes, bolts, nuts, screws, cables, valves etc., in excess quantities than required. This is in order to avoid shortages during the assembly due to damage or any other cause as it is difficult to procure the accessories locally. It is notified that all the shipped accessories during assembly and the excess accessories after commissioning are always remain the properties of WTT. WTT has the authority to retake the same at any period of time, even before and after commissioning. It is client's responsibility to preserve the above-mentioned items till WTT takes back the same.

13.3. START-UP AND COMMISSIONING

Installation & removal of temporary screens on all process lines Flushing and disinfection of all piping and tanks (including process equipment tanks) and verification of removal of all residual debris from construction. Alignments & required materials for rotating equipment MEG testing of all field motor power wiring (as required). Continuity checks for all electrical field wiring as per Installation Checklist, Hydro-testing of all field installed piping.

Supply raw materials, oil/lubricants chemicals and utilities during start-up and operation. Electrical & Mechanical support labor for commissioning activities, loading of membranes, stacks, modules, etc.

13.4. FACILITY MANAGEMENT

Client will provide such warehouse storage space and facilities, as are available at the site, and are reasonably appropriate to store parts, consumables, tools, etc. in accordance with manufacturer's recommendations. Such warehouse storage space will be a segregated area, secured and protected from adverse climate as may reasonably be required. The storage area shall be facilitated with 24 hours lock & key with security and Client will be responsible for risk of loss of WTT's parts while in storage at the site. Client will maintain WTT's parts stored at the site free & clear of any and all liens of Client and lenders, bondholders, contractors & other creditors of any nature.

Client will afford WTT's personnel free access and egress of the facility for all authorized work. Provide workshop facilities/area with roof and stabilized power supply, as is reasonably appropriate to carry out machining/fabrication works.

Provide adequate illumination and emergency lighting for all areas in which the WTT will be executing the scope of supply. Identify a Client project contact person to be available to WTT's personnel to address any issues related to WTT's execution of WTT's scope of work. Responsible for the equipment for movement of chemical drums, totes, and resins, as is reasonable. Provide all site utilities such as raw water, instrument quality air, potable water and power required for operation of the proposed equipment included in this scope of supply.

13.5. CONDITIONAL OFFERING

Client understands that this order confirmation has been issued based upon the information provided by Client and currently available to, WTT at the time of proposal issuance. Any changes or discrepancies in site conditions (including but not limited to system influent characteristics, changes in Environmental Health and Safety ("EH&S") conditions, and/or newly discovered EH&S concerns), Client financial standing, Client requirements, or any other relevant change, or discrepancy in, the factual basis upon which this proposal was created, may lead to changes in the offering, including but not limited to

changes in pricing, warranties, quoted specifications, or terms and conditions. WTT's offering in this proposal is conditioned upon a full WTT EH&S.

14. APPENDIX

14.1. APPENDIX A: CLARIFICATIONS

- Pump MOC has been considered as per WTT standard practice.
- The Equipment in WTT scope will be procured as per WTT Standard Vendor List.
- WTT has not envisaged third party equipment inspection before dispatch.
- The payments received once could not be processed again for refund.
- WTT has considered instruments as per WTT standard practice
- Painting specifications of WTT supplied equipment is as per standard manufacturer painting specifications.

14.2. APPENDIX B: ACCEPTANCE

To ensure accurate and prompt order entry, product delivery, billing and accounts receivables processing, please ensure your Purchase Order contains the complete Client and order details.

Please communicate your PO through:

E-mail	venkat@wttint.com
Postal Mail: (Indian Office)	WTT INTERNATIONAL PVT. LTD. No. 3, College cross road Avarangadu, Tiruppur, TamilNadu, INDIA – 641602

Bank Account details:

Beneficiary Name : WTT INTERNATIONAL PVT. LTD.,

Bank Name : INDIAN BANK LTD

Branch Name : TIRUPUR MSME BRANCH, TIRUPUR, TAMILNADU.

Account Number OD : 6475173650,

IFSC Code : IDIB000I043

14.3. APPENDIX C: WARRANTY

The mechanical warranty is only applicable to equipment supplied by WTT. The mechanical warranty period on all equipment supplied, unless otherwise noted, is twelve (12) months from the date of installation or fourteen (14) months from equipment shipment, whichever occurs first. WTT's obligation under this warranty is to the repair or replace, of any device or part thereof, which shall prove to have been manufacturing defects.

This warranty excludes the diffuser membranes, electrical items, defects, failures, damages or performance limitations caused in whole or in part by normal wear & tear, power failures, surges, fires, floods, snow, ice, lightning, excessive heat or cold, highly corrosive environments, accidents, actions of third parties, or other events outside of WTT's control. Warranty period for the entire equipment including replaced or repaired parts will be limited to the unexpired portion of the total warranty period. Bought out components are guaranteed only to the extent of guarantees given to us by our suppliers.

WTT assumes no liability for any damage to equipment caused by inadequate storage or handling as per manufacturer's recommendations in supplied technical literature, or by defective or sub-standard workmanship or materials provided by Client or any other third party responsible for handling, storing or installing the equipment. Client undertakes to give immediate notice to WTT if goods or performance appear defective and to provide WTT with reasonable time and opportunity to make inspections and tests. If WTT is not at fault, Client shall pay WTT the costs and expenses of the inspections and tests.

Goods shall not be returned to WTT without WTT's permission. WTT will provide Client with a "Return Goods Authorization" (RGA) number to use for returned goods. All return costs associated with shipping and labor are not included in the mechanical warranty. WTT warrants, subject to the provisions herein after set forth, that after stable operation of the WTT system has been attained and operators have acquired reasonable skills, the Equipment supplied for this project will be capable of producing the results set forth in stage wise parameter table, provided that:

The Equipment is operated and maintained at all times in accordance with the WTT Operations and Maintenance manual, The Equipment is operated within the mixed liquor characteristics defined in Influent quality table of this section, WTT has, until performance of its obligation herein is met, reasonable access to the Equipment and the operational data

relating thereto, Client furnishes adequate and competent operating, supervisory and maintenance staff, and necessary laboratory facilities with test equipment and personnel, Client utilizes the services of WTT until its performance obligations are met, Client supplies all necessary raw materials and services of a quantity and of a quality specified by WTT, An adequate and continuous power supply is available that will enable operation of all required equipment.